

CS 506: Introduction to Quantum Computing

Phase Kickback (continued) and Deutsch's Algorithm

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1 Overview

This lecture closes off phase kickback and introduces Deutsch's problem, which is solved by Deutsch and Deutsch-Jozsa algorithms. While exploring the phase kickback, we generalized the control-gate, calling it the U_f gate. In the second half, we explored Deutsch's problem and a modern version of its solution. Lastly, we introduced Deutsch-Jozsa algorithm which extends Deutsch's algorithm to n bits.

2 Phase Kickback (continued from last lecture)

Reminder: Control-NOT (CNOT) gate

For $b \in \{0, 1\}$,

$$\text{CNOT} |b\rangle \left(\frac{|0\rangle - |1\rangle}{\sqrt{2}} \right) = (-1)^b |b\rangle \left(\frac{|0\rangle - |1\rangle}{\sqrt{2}} \right).$$

The state following state is the most important part of the phase-kickback:

$$\frac{|0\rangle - |1\rangle}{\sqrt{2}}$$

The eigenvalue of the CNOT:

$$(-1)^b$$

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Continuation: We can generalize this controlled gate as U_f (Control- f gate)

For

$$f : \{0, 1\} \rightarrow \{0, 1\}.$$

$$U_f |x\rangle |y\rangle = |x\rangle |y \oplus f(x)\rangle, \quad x, y \in \{0, 1\}.$$

$$\begin{cases} \text{if } x = |0\rangle, & U_f |0\rangle |y\rangle = |0\rangle |y\rangle, \\ \text{if } x = |1\rangle, & U_f |1\rangle |y\rangle = |1\rangle |f(y)\rangle. \end{cases}$$

Identifying the phase kickback

To see how the phase kickback shows up mathematically, we can solve U_f for its eigenvalues.

$$U_f |x\rangle |y\rangle = |x\rangle |y \oplus f(x)\rangle, \quad x, y \in \{0, 1\}.$$

Consider:

$$U_f |x\rangle \left(\frac{|0\rangle - |1\rangle}{\sqrt{2}} \right)$$

The following operations are to find the eigenvalues:

$$\begin{aligned} U_f |x\rangle \left(\frac{|0\rangle - |1\rangle}{\sqrt{2}} \right) &= \frac{U_f |x\rangle |0\rangle - U_f |x\rangle |1\rangle}{\sqrt{2}} \\ &= \frac{|x\rangle |0 \oplus f(x)\rangle - |x\rangle |1 \oplus f(x)\rangle}{\sqrt{2}} \\ &= |x\rangle \left(\frac{|0 \oplus f(x)\rangle - |1 \oplus f(x)\rangle}{\sqrt{2}} \right). \end{aligned}$$

Case by case:

If $f(x) = 0$,

$$|x\rangle \left(\frac{|0 \oplus 0\rangle - |1 \oplus 0\rangle}{\sqrt{2}} \right) = |x\rangle \left(\frac{|0\rangle - |1\rangle}{\sqrt{2}} \right).$$

If $f(x) = 1$,

$$|x\rangle \left(\frac{|0 \oplus 1\rangle - |1 \oplus 1\rangle}{\sqrt{2}} \right) = |x\rangle \left(\frac{|1\rangle - |0\rangle}{\sqrt{2}} \right) = -|x\rangle \left(\frac{|0\rangle - |1\rangle}{\sqrt{2}} \right).$$

Combining both together:

$$U_f |x\rangle \left(\frac{|0\rangle - |1\rangle}{\sqrt{2}} \right) = (-1)^{f(x)} |x\rangle \left(\frac{|0\rangle - |1\rangle}{\sqrt{2}} \right).$$

The 2 eigenvalues are:

$$(-1)^{f(x)}, \quad x \in \{0, 1\}.$$

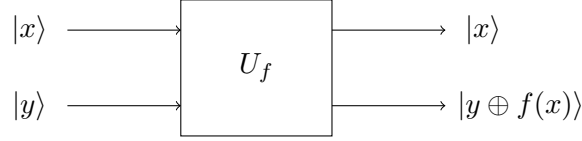
This phase kickback is here: $(-1)^{f(x)}$

The first qubit is represented by: $(-1)^{f(x)} |x\rangle$

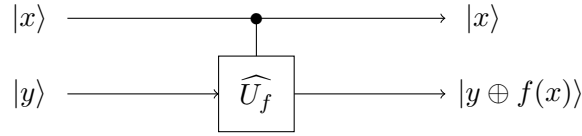
Generally:

$$U_f (\alpha_0 |0\rangle + \alpha_1 |1\rangle) \left(\frac{|0\rangle - |1\rangle}{\sqrt{2}} \right) = \left((-1)^{f(x)} \alpha_0 |0\rangle + (-1)^{f(x)} \alpha_1 |1\rangle \right) \left(\frac{|0\rangle - |1\rangle}{\sqrt{2}} \right).$$

Diagrams for U_f :



or



for:

$$\widehat{U}_f |y\rangle = |y \oplus f(x)\rangle$$

Deutsch Problem (Improved Solution)

Given a function

$$f : \{0, 1\} \rightarrow \{0, 1\}$$

We can query: giving $x \in \{0, 1\}$ as input and get $f(x)$ as output

Goal: Figure out if

$$f(0) = f(1) \rightarrow \text{constant}$$

or

$$f(0) \neq f(1) \rightarrow \text{balanced}$$

Classical algorithms need two queries.

A quantum algorithm needs one query: Quantum input to U_f then obtain output of U_f

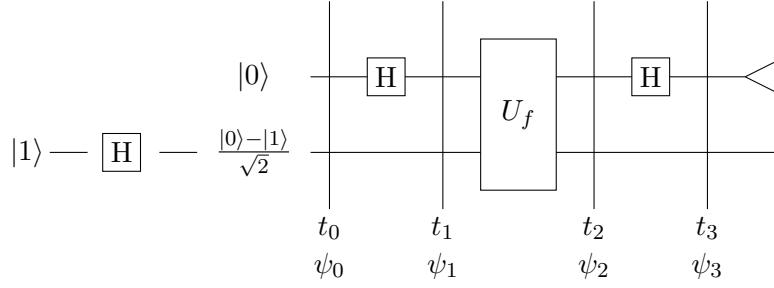
Wrong solution

$$\begin{aligned} U_f \left(\frac{|0\rangle + |1\rangle}{\sqrt{2}} \right) |0\rangle &= \frac{U_f |0\rangle |0\rangle + U_f |1\rangle |0\rangle}{\sqrt{2}} \\ &= \frac{|0\rangle |0 \oplus f(1)\rangle + |1\rangle |0 \oplus f(0)\rangle}{\sqrt{2}} \\ &= \frac{|0\rangle |f(0)\rangle}{\sqrt{2}} + \frac{|1\rangle |f(1)\rangle}{\sqrt{2}} \end{aligned}$$

This doesn't work because obtaining one qubit will destroy the other.

Correct solution

The circuit as follows:



At t_0 ,

$$|\psi_0\rangle = |0\rangle \left(\frac{|0\rangle - |1\rangle}{\sqrt{2}} \right)$$

At t_1 ,

$$\begin{aligned} |\psi_1\rangle &= H |0\rangle \left(\frac{|0\rangle - |1\rangle}{\sqrt{2}} \right) \\ &= \left(\frac{|0\rangle + |1\rangle}{\sqrt{2}} \right) + \left(\frac{|0\rangle - |1\rangle}{\sqrt{2}} \right) \\ &= \frac{1}{\sqrt{2}} |0\rangle \left(\frac{|0\rangle - |1\rangle}{\sqrt{2}} \right) + \frac{1}{\sqrt{2}} |1\rangle \left(\frac{|0\rangle - |1\rangle}{\sqrt{2}} \right) \end{aligned}$$

At t_2 ,

$$\begin{aligned} |\psi_2\rangle &= U_f |\psi_1\rangle \\ &= \frac{(-1)^{f(0)}}{\sqrt{2}} |0\rangle \left(\frac{|0\rangle - |1\rangle}{\sqrt{2}} \right) + \frac{(-1)^{f(1)}}{\sqrt{2}} |1\rangle \left(\frac{|0\rangle - |1\rangle}{\sqrt{2}} \right) \\ &= \left(\frac{(-1)^{f(0)} |0\rangle + (-1)^{f(1)} |1\rangle}{\sqrt{2}} \right) \left(\frac{|0\rangle - |1\rangle}{\sqrt{2}} \right) \\ &= (-1)^{f(0)} \left(\frac{|0\rangle + (-1)^{f(1)-f(0)} |1\rangle}{\sqrt{2}} \right) \left(\frac{|0\rangle - |1\rangle}{\sqrt{2}} \right) \end{aligned}$$

There are two cases.

Case 1: $f(0) = f(1)$

Then

$$f(0) \oplus f(1) = 0,$$

so

$$|\psi_2\rangle = (-1)^{f(0)} \left(\frac{|0\rangle + |1\rangle}{\sqrt{2}} \right) \left(\frac{|0\rangle - |1\rangle}{\sqrt{2}} \right).$$

Case 2: $f(0) \neq f(1)$

Then

$$f(0) \oplus f(1) = 1,$$

so

$$|\psi_2\rangle = (-1)^{f(0)} \left(\frac{|0\rangle - |1\rangle}{\sqrt{2}} \right) \left(\frac{|0\rangle - |1\rangle}{\sqrt{2}} \right).$$

At t_3 ,

Case 1: $f(0) = f(1)$

$$\begin{aligned} |\psi_3\rangle &= (-1)^{f(0)} H \left(\frac{|0\rangle + |1\rangle}{\sqrt{2}} \right) \left(\frac{|0\rangle - |1\rangle}{\sqrt{2}} \right) \\ &= (-1)^{f(0)} |0\rangle \left(\frac{|0\rangle - |1\rangle}{\sqrt{2}} \right). \end{aligned}$$

Observe $|0\rangle$ with probability

$$\left((-1)^{f(0)} \right)^2 = 1.$$

Case 2: $f(0) \neq f(1)$

$$\begin{aligned} |\psi_3\rangle &= (-1)^{f(0)} H \left(\frac{|0\rangle - |1\rangle}{\sqrt{2}} \right) \left(\frac{|0\rangle - |1\rangle}{\sqrt{2}} \right) \\ &= (-1)^{f(0)} |1\rangle \left(\frac{|0\rangle - |1\rangle}{\sqrt{2}} \right). \end{aligned}$$

Observe $|1\rangle$ with probability

$$\left((-1)^{f(0)} \right)^2 = 1.$$

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Deutsch's Original Solution

Deutsch's original solution is left as an exercise for students. Here are some notes about it:

1. Start with $|0\rangle|0\rangle$ instead of

$$|0\rangle \left(\frac{|0\rangle - |1\rangle}{\sqrt{2}} \right).$$

2. Can get the answer with probability of $3/4$.
3. Can determine whether the answer is correct or not with probability of $1/2$.

Overview

Deutsch Problem

Given a black-box function

$$f : \{0, 1\} \rightarrow \{0, 1\},$$

determine whether

$$f(0) = f(1) \rightarrow \text{constant}$$

or

$$f(0) \neq f(1) \rightarrow \text{balanced}.$$

The goal is to determine this using queries to f .

Deutsch-Jozsa Problem

Given

$$f : \{0, 1\}^n \rightarrow \{0, 1\},$$

determine if $f(x)$ values are all the same for

$$x \in \{0, 1\}^n,$$

where f is guaranteed to be either constant or balanced.

$$\frac{1}{2}x2^n = 2^{n-1}$$

Or

$$f(x) = 0 \quad \text{for 50\% of inputs,}$$

$$f(x) = 1 \quad \text{for 50\% of inputs.}$$

Classical queries required:

$$\frac{1}{2}x2^n = 2^{n-1} + 1$$

While quantum only needs 1 qubit.

To be continued in the next lecture.